

Spatial and temporal aquatic–terrestrial transitions in the temporary Albarine River, France: responses of invertebrates to experimental rewetting

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SUMMARY

1. In temporary rivers, viewed as coupled terrestrial–aquatic ecosystems, spatial and temporal transition zones between aquatic and terrestrial conditions are common and occur simultaneously.
2. The effects of artificial rewetting on terrestrial and aquatic invertebrate assemblages were examined in dry sediments collected from the Albarine River, France. Rewetted sediments had previously been dry for between 0.1 and 142 days. Dry sediments were collected directly from the streambed (DS) and from riparian gravel bars (RGB).
3. We first predicted that invertebrate responses to rewetting would vary with the duration of the preceding dry period. Second, we predicted convergence of the invertebrate assemblages in DS and RGB sediments with increasing duration of the dry period. Third, we predicted that an aquatic ‘invertebrate seedbank’ (aquatic life stages that persist within streambed sediments during dry periods) would contribute substantially to the resilience of benthic assemblages.
4. Results indicated that the duration of the dry period was the primary driver of aquatic and terrestrial responses to artificial rewetting. The density and richness of aquatic taxa decreased with the duration of the dry period in both DS and RGB sediments, whereas the density of terrestrial invertebrates increased in DS sediments.
5. No convergence between DS and RGB assemblage composition was observed with an increasing dry period. Although there were more aquatic organisms in DS sediments than in RGB sediments, there was no difference in taxonomic richness between sediment types. Even after prolonged dry periods (142 days), there was typically a lower density and taxonomic richness of terrestrial invertebrates in DS sediments than in adjacent RGB sediments.
6. The results suggest that the aquatic invertebrate seedbank could contribute substantially to the resilience of benthic assemblages in the Albarine River, in addition to other mechanisms such as drift and oviposition. Of the taxa in the benthos before and after the summer dry period, 65% were also recovered from artificially rewetted DS sediments. The simultaneous presence of temporal and spatial terrestrial–aquatic transition zones in temporary rivers increases successional diversity (i.e. mosaics of dry and saturated streambed patches at various stages of terrestrial and aquatic succession). This contribution to biodiversity emphasises the need to protect dry reaches of temporary rivers.

Keywords: aestivation, dry events, ecotones, flow intermittence, resilience

Introduction

Ecotones have been defined spatially as narrow transition zones between dissimilar environments (Naiman & Decamps, 1997), and temporally as short transitional

periods between persistent and dissimilar states (Pinay *et al.*, 1990; Larned, Datry & Robinson, 2007). Processes occurring along these transition zones and over varying temporal resolutions influence the long-term dynamics of communities and ecosystems (Naiman, Decamps &

Pollock, 1993; Ward, Tockner & Schiemer, 1999). In temporary rivers, which can be viewed as coupled terrestrial-aquatic ecosystems (Larned *et al.*, 2010), spatial and temporal transition zones between aquatic and terrestrial conditions are common and occur simultaneously.

In natural rivers, fringing riparian zones (e.g. protruding gravel bars and banks) constitute spatial transition zones between aquatic and terrestrial conditions and have high biodiversity (Gregory *et al.*, 1991; Naiman & Decamps, 1997; Ward & Tockner, 2001; Tockner & Stanford, 2002; Paetzold, Schubert & Tockner, 2005). In particular, riparian zones are inhabited by a diverse terrestrial invertebrate fauna including spiders, rove beetles, ground beetles and ants (Hering & Plachter, 1997; Manderbach & Hering, 2001; Sadler, Bell & Fowles, 2004; Paetzold *et al.*, 2005). Several of these taxa are predatory (e.g. lycosid spiders, carabid beetles), and their diet includes aquatic invertebrates inhabiting the adjacent stream channel (Hering & Plachter, 1997; Paetzold *et al.*, 2005; Lamberts *et al.*, 2008; Greenwood & McIntosh, 2010). During dry phases in temporary streams, riparian zones gradually become part of the dry riverbed. As a consequence, the terrestrial predators inhabiting riparian gravel bars may decrease as the duration of the dry period increases.

Alternating wet and dry periods in temporary rivers constitute temporal transition zones (Larned *et al.*, 2010; Datry *et al.*, 2011). When surface water disappears, riverbed sediments shift rapidly from aquatic to terrestrial conditions. Consequently, the invertebrate assemblage shifts from one characteristic of flowing waters to one typical of a dry riverbed (Larned *et al.*, 2007). These changes relate first to taxon-specific responses to desiccation. Some desiccation-sensitive aquatic taxa die within a few hours of the sediment drying if they cannot find an aquatic refuge (e.g. perennial section, hyporheic zone, lateral pond). Other aquatic taxa can tolerate a dry period of a few days if the streambed sediments remain relatively moist (Stanley *et al.*, 1994), while some aquatic taxa (e.g. some Chironomidae, Copepoda, Cladocera) can persist in dry sediments for several months as desiccation-resistant forms (Harper & Hynes, 1970; Williams & Hynes, 1976; Boulton, 1989; Tronstad, Tronstad & Benke, 2005). As a result, the duration of the dry period in temporary rivers may have a considerable influence on the aquatic invertebrate assemblage (e.g. overall density, diversity and structure) and its response to subsequent inundation (Larned *et al.*, 2007).

Aquatic invertebrates persisting in streambed sediments during dry phases have been proposed to form an 'invertebrate seedbank', which may facilitate the

recovery of the benthos after rewetting (Boulton, 1989; Tronstad *et al.*, 2005; Larned *et al.*, 2007; Stubbington *et al.*, 2009). However, the contribution of invertebrate seedbanks to the persistence of the benthos in temporary rivers has not yet been quantified. During aquatic-terrestrial transition periods, invertebrate assemblages may also shift in composition as a consequence of the colonisation of dry riverbeds by terrestrial invertebrates from adjacent fringing habitats (Wishart, 2000; Greenwood & McIntosh, 2010; Corti *et al.*, 2011; Steward *et al.*, 2011). Although terrestrial successions in dry riverbeds have not yet been described, invertebrate assemblages from dry riverbeds and lateral riparian zones may converge over time.

We examined the effects on terrestrial and aquatic invertebrate assemblages of experimental rewetting of dry sediments collected from 13 sites located along a 15-km temporary reach of the Albarine River, France. Sites along the study reach had previously been dry for between 0.1 and 142 days. Dry sediments were collected from the dry streambed (DS) and from riparian gravel bars (RGB). We first predicted that invertebrate responses to rewetting would vary with the duration of the preceding dry period. Second, we predicted a convergence of invertebrate assemblage composition from rewetted DS and RGB sediments with increasing duration of the dry period. Third, if there is a significant contribution of the invertebrate seedbank to recovery upon rewetting, we predicted that invertebrate assemblages from rewetted DS sediments and benthic assemblages before and after dry periods would have many taxa in common.

Methods

Study sites

The Albarine River drains a 313-km² catchment in temperate eastern France (Fig. 1). For a 15-km stretch, the downstream section of the river flows across an alluvial plain underlain by thick (>36 m) glacio-fluvial deposits, the permeability of which ranges from 2×10^{-2} to 7×10^{-2} m s⁻¹ (Datry *et al.*, 2011). Along this alluvial plain, the river continuously infiltrates into the underlying aquifer, which gradually leads to complete river drying (Fig. 1). Drying typically starts in spring at the confluence with the Ain River, and the drying front then progressively moves upstream throughout the summer. Flow generally resumes in late autumn/early winter (Datry, 2012; Datry *et al.*, 2011; Larned *et al.*, 2011). In 2010, the onset of drying started in mid-May at the confluence (Fig. 1), and the complete rewetting of the river occurred in late October.

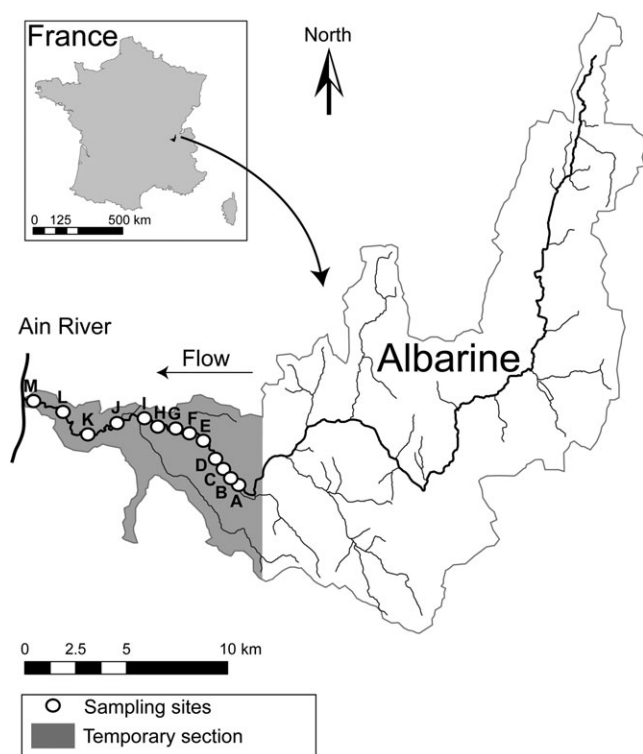


Fig. 1 Map of the Albarine River catchment, France, showing perennial and temporary sections. Letters along the main stem indicate sampling locations.

Thirteen sites (A–M) with different dry periods were selected along the temporary section of the river (Fig. 1). The dry period at each site was estimated at the time of sampling from the time elapsed since the last observed flow (on 14 May 2010). Between the onset of drying at site M (which dried first, A drying last) and the 20 September 2010, sites were visited every 1–10 days to allow an estimation of the dry period to the nearest 10 days. Sites G–M were sampled on 20 September 2010. The drying reach was then visited daily to estimate the dry period duration to within one hour. Sites A to F were sampled on 10 October 2010. As a result, site A was just drying (1.5 h, 0.1 day), whereas site M had been dry for 142 days at the time of sampling; the remaining 11 sites were positioned along this gradient of dry period duration (Table 1).

Invertebrate responses to experimental rewetting

At each site, two or three dry sediment samples were collected from random points in riffles within the dry streambed (DS) (Table 1). Except at sites C, D, E and F, two dry sediment samples were also collected at the same time from random points from one adjacent dry RGB (Table 1). Gravel bars were located 2–4 m laterally from

Table 1 Sites, duration of the dry period and number of sediment samples collected for artificial rewetting and sediment moisture content

Site	Dry period duration (days)	Sediment samples for rewetting		Sediment samples for % moisture	
		DS	RGB	DS	RGB
A	0.1	3	2	2	2
B	0.5	3	2	2	2
C	0.9	3	x	2	2
D	2	3	x	2	2
E	5	3	x	2	2
F	7	3	x	2	2
G	9	3	2	2	2
H	25	3	2	2	2
I	40	2	2	2	2
J	50	2	2	2	2
K	100	2	2	2	2
L	130	2	2	2	2
M	142	2	2	2	2
Total		34	18	26	26

DS, dry streambed; RGB, riparian gravel bars.

the river at mean flow conditions and had little vegetation at the time of sample collection. No differences in grain size were apparent between sediments from the riverbed and adjacent gravel bars. Each sample consisted of 2.5 L of sediment, which was manually excavated from a 0.2-m² surface area to a depth of about 10 cm and then placed in an 8-L container. In addition, two additional 1-L sediment samples were collected from the DS and RGB sampling areas, transferred to plastic bags and sealed for the determination of sediment moisture. All samples were transported to the laboratory within 4 hr. Samples for the determination of sediment moisture content were immediately frozen and processed within 4 weeks. The 8-L containers were placed immediately in an environmental chamber (12 h:12 h light/dark cycle, 20 °C). Each container was flooded with 5 L of filtered, non-chlorinated tap water, continuously aerated using airstones, and fitted with 1-mm mesh lids to retain emerging insects. Immediately after water was added, and subsequently at $t = 4, 8, 16$ and 32 days, the water column of each container was swept vigorously using a 200- μ m hand-net for 30 s to collect living aquatic and dead terrestrial invertebrates. Material retained on the mesh was preserved in 96% ethanol. A wet period of 32 days was selected as a compromise between a long duration that could result in sediment anoxia and one too brief for eggs and cysts to hatch (Larned *et al.*, 2007). After 32 days, the sediments in each container were vigorously elutriated six times and each elutriate poured through a

200- μ m mesh sieve; retained material was preserved in 96% ethanol.

Sediment moisture content

For each 1-L sediment sample, two subsamples of c. 50 mL were weighed wet, dried at 60 °C to constant mass and reweighed. Moisture content was determined from differences between wet and dry mass.

The benthos

At five sites (E, F, H, K and M), samples of benthic invertebrates were collected before and after drying events, for comparison with invertebrate assemblages from experimentally rewetted sediments. Benthic samples before drying were collected in early May 2010, 13 days before the section started to dry, and post-drying benthic samples were collected in November 2010, 21 days after the entire section rewetted. At each site, four benthic samples were collected at random from two riffles. Invertebrates were sampled using a Hess sampler (200- μ m mesh, 0.125 m² in diameter, down to a 10 cm depth) and were preserved in 96% ethanol.

Invertebrates were identified to the lowest taxonomic level possible, which varied with taxonomic group and developmental stage. Most aquatic insects and molluscs were determined to genus, Chironomidae and most terrestrial insects were identified to the family level and other taxa to the family, order or class. Taxa were classified as aquatic, semi-aquatic, aerial stages of aquatic taxa or terrestrial (Tachet *et al.*, 2000). Semi-aquatic taxa included those for which a definite assignment to aquatic or terrestrial classes could not be made, mainly because of the low taxonomic resolution (e.g. Acarina, Oligochaeta and Nematoda).

Data analysis

Faunal differences between dry streambed and riparian gravel bar sediments. Invertebrates recovered from the water column and from the sediments during and at the end of the rewetting experiment were pooled together. Nonparametric Wilcoxon sum-rank tests were used to compare taxonomic richness and total density for all invertebrate classes (i.e. aquatic, semi-aquatic, aerial stages of aquatic taxa and terrestrial), and sediment moisture content between DS and RGB sediments.

Effects of the length of the dry period on invertebrate assemblages. Linear and nonlinear regressions were used

to examine relationships between the duration of the preceding dry period and taxonomic richness and density for all invertebrate classes and sediment moisture content of DS and RGB sediments. Compositional changes in DS and RGB invertebrate assemblages with an increasing dry period were examined using non-metric multidimensional scaling (NMDS). Similarity matrices were calculated using Bray–Curtis distances based on log-transformed density data. Both for DS and RGB, nestedness analysis was used to examine whether the invertebrate taxa (all classes considered) collected at sites with a long preceding dry period duration were nested subsets of taxa found at sites with a shorter preceding dry period. A detailed explanation of nestedness analysis is given in Atmar & Patterson (1993). Briefly, nestedness analyses operate on matrices of sites versus taxon presence and absence data. A nestedness calculator reorders the rows and columns of the matrix until the cells with taxa present are concentrated into one corner of the matrix (i.e. the matrix is ‘packed’ with as few empty cells as possible). An index of nestedness (*T*, temperature) is then calculated from a comparison of the observed (packed) matrix with a maximally nested matrix. Monte Carlo permutation tests are used to generate random matrices, and the probability that the observed nestedness has occurred by chance is determined by comparing the observed *T* with *T*-values from the random matrices. Bray–Curtis similarity between DS and RGB sediments was calculated based on log-transformed density data from the nine sites (A, B, G, H, I, J, K, L and M) where both sample types were collected, and examined along the gradient of increasing dry period duration.

Experimentally rewetted sediment and benthic invertebrates. The occurrence of each taxon among the different sample types (pre- and post-drying benthic invertebrate samples and rewetted DS sediment samples) was examined for the five sites (E, F, H, K and M) at which all sample types were collected. Bray–Curtis similarities were then calculated to examine assemblage similarities among the three sample types. Invertebrate assemblages could not be compared quantitatively, as sampling techniques differed, so similarities were calculated based on presence–absence data. This analysis was conducted for all taxa occurring simultaneously in the three data sets and then terrestrial taxa removed to focus on aquatic and semi-aquatic taxa.

All statistical analyses were performed using R software (R Development Core Team, 2011). The significance level for all statistical analyses was set at $P = 0.01$.

Results

Fauna of the dry streambed and riparian gravel bar sediments

A total of 21 539 individuals belonging to 73 taxa were recovered from experimentally rewetted sediments (Appendix S1). From DS sediments, 18 470 individuals (72 taxa) were collected, of which 86% were aquatic (46 taxa), 8% semi-aquatic (seven taxa), 6% aerial stages of aquatic taxa (four taxa) and <1% terrestrial individuals (18 taxa). From RGB sediments, 2747 individuals (52 taxa) were collected, of which 33% were aquatic (28 taxa), 56% semi-aquatic (seven taxa), 1% aerial stages of aquatic taxa (seven taxa) and 10% terrestrial individuals (15 taxa). The two most abundant aquatic taxa were Cladocera (59.1%) and Cyclopoida (9.2%) for DS, and Cladocera (13.2%) and Ostracoda (11.6%) for RGB sediments. Oligochaeta was the most abundant semi-aquatic taxon in both DS (6.7%) and RGB (33.5%) sediments. Adult Chironomidae were the most abundant aerial stages of aquatic taxa in DS (4.3%) and RGB (1.6%) sediments. Hemiptera was the most abundant terrestrial taxon in DS sediments (0.2%), whereas Thysanoptera (0.6%) was the most abundant terrestrial taxon in RGB sediments.

There were no differences between DS and RGB sediments in the taxon richness of total, aquatic, semi-aquatic or aerial stages of aquatic taxa. In contrast, the number of terrestrial taxa was higher in RGB than in DS sediments (Table 2, Wilcoxon sum-rank test, $P = 0.006$). The overall densities of total and aquatic taxa were higher on DS than on RGB sediments (Table 2, Wilcoxon sum-rank test, $P = 0.003$). The density of aerial stages of aquatic taxa did not differ between sample types. Semi-aquatic and terrestrial densities were higher in RGB than in DS

sediments (Table 2, Wilcoxon sum-rank test, $P = 0.008$ and 0.001 , respectively). Sediment moisture content was higher in RGB than in DS sediments (10.5 ± 4 and $6.5 \pm 2\%$, respectively, Wilcoxon sum-rank test, $P = 0.002$).

Duration of the dry period

Dry streambed sediments. Aquatic taxon richness decreased steadily along the gradient of increasing dry period duration (Fig. 2a). On average, for every 10-day increase in the dry period, approximately one aquatic taxon was lost from DS sediments. Semi-aquatic, aerial stages of aquatic taxa and terrestrial taxon richness did not show any relationship with length of the dry period (Fig. 2a). In contrast, the density of aquatic taxa declined as a power function of increasing duration of the dry period (Fig. 2a). An increase in the duration of the dry period from 0.1 to 10 days resulted in a 20-fold decrease in the density of aquatic invertebrates. For preceding dry periods from 7 to 142 days, there was no substantial change in the density of aquatic taxa, which remained below 50 organisms 2.5 L^{-1} . Terrestrial density increased steadily with length of the dry period (Fig. 2a). However, after 142 days of drying, mean terrestrial density on DS sediments was only 2 ± 1 organisms 2.5 L^{-1} . Density of semi-aquatic taxa and aerial stages of aquatic taxa did not change with increasing dry period duration.

The NMDS plot showed a clear separation of sites according to the duration of dry period, from A on the left-hand side to M on the right-hand side (Fig. 3a). The densities of aquatic taxa, such as Cyclopoida, Harpacticoida, Cladocera and Diptera larvae (e.g. Chironomidae, Orthocladinae and Tanypodinae), Ephemeroptera (e.g.

Table 2 Minimum, maximum and mean ($\pm$ SD) of total, aquatic, semi-aquatic and aerial stages of aquatic taxa and terrestrial richness and density recovered in rewetted sediments from the dry streambed (DS) and riparian gravel bars (RGB)

	DS			RGB		
	Min.	Max.	Mean $\pm$ SD	Min.	Max.	Mean $\pm$ SD
Total richness	15	39	23.9 ± 7.4	12	27	19.4 ± 4.7
Aquatic richness	6	28	15.3 ± 6.3	5	19	9.9 ± 4.9
Semi-aquatic richness	2	5	2.9 ± 1.0	2	5	3.1 ± 0.9
Aerial richness	0	3	1.5 ± 0.8	0	2	1.0 ± 0.7
Terrestrial richness	0	3	1.2 ± 1.2	1	8	3.8 ± 2.7
Total density	44	4532	980.7 ± 1382.3	29	482	214.4 ± 140.2
Aquatic density	15	4439	918.2 ± 1382.1	11	424	113.5 ± 128.7
Semi-aquatic density	20	72	35.1 ± 16.2	29	147	70.4 ± 37.7
Aerial density	0	4	1.4 ± 1.2	0	4	1.1 ± 1.2
Terrestrial density	0	2.5	1 ± 1	2	148	21.9 ± 47.2

Values are given for 2.5 L of sediments.

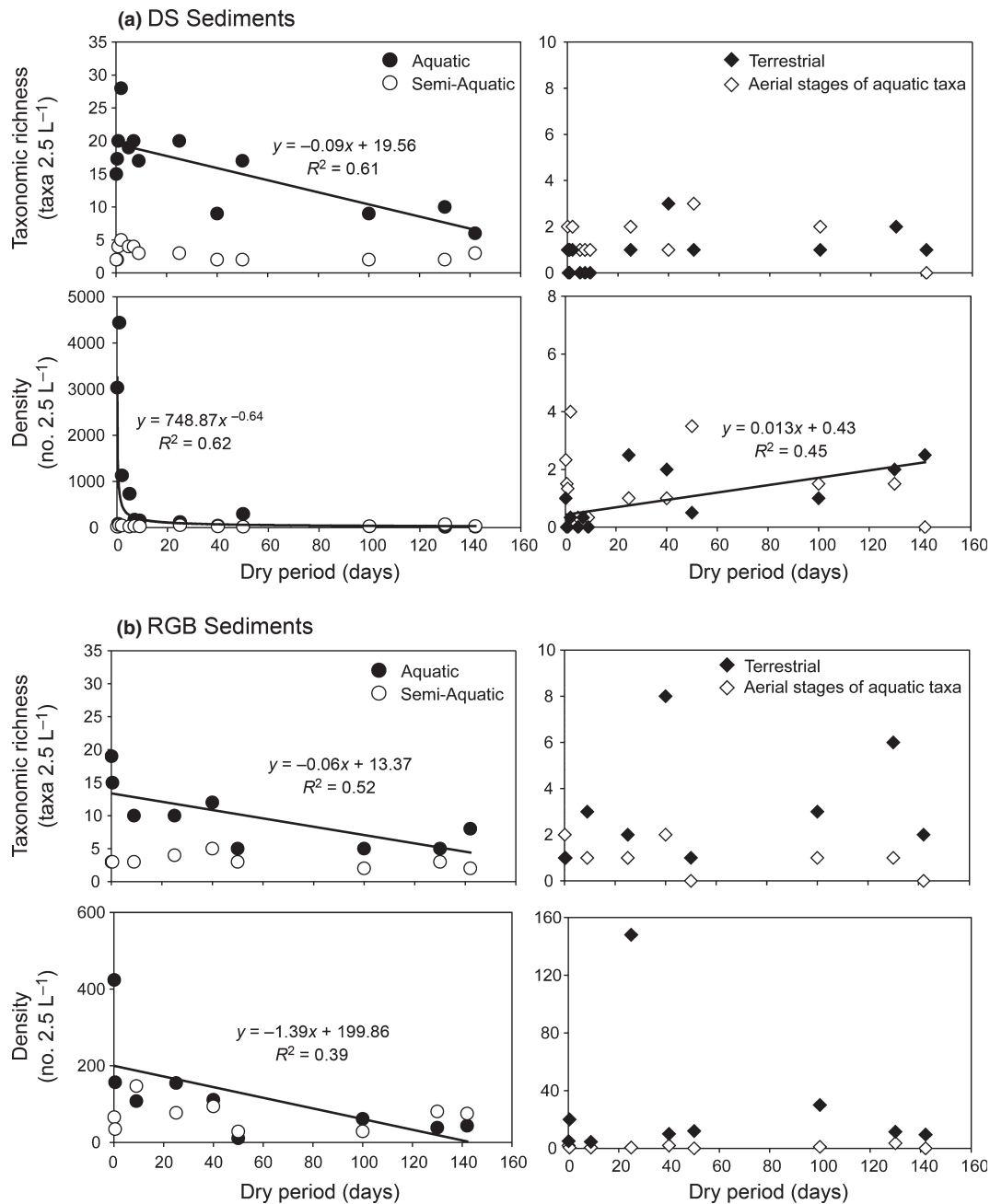


Fig. 2 Relationships between duration of the dry period and taxonomic richness and density of aquatic, semi-aquatic, aerial stages of aquatic taxa and terrestrial taxa from (a) dry streambed (DS) and (b) riparian gravel bar (RGB) sediments.

Baetis spp.) and Coleoptera (e.g. *Esolus* spp.), were high at sites with a preceding dry period <1 week (A, B, C, D and E) (Fig. 3a). Sediments from sites with a preceding dry period between 7 and 40 days (F, G, H) had a high density of aquatic taxa, such as larvae of Diptera (e.g. Ceratopogonidae), Ephemeroptera (e.g. Baetidae; *Centroptilum luteolum*, Müller, 1776; *Habroleptoides* spp.), Coleoptera (e.g. *Oulimnius* spp.) and Plecoptera (*Leuctra* spp.), and semi-aquatic taxa such as Acarina (Fig. 3a). Sedi-

ments from sites with a long preceding dry period (>40 days; I, J, K, L, M) had a high density of aquatic taxa, such as Ostracoda or Coleoptera (adult *Esolus* spp.), a high density of semi-aquatic taxa such as Collembola, Nematoda and Oligochaeta, and a high density of terrestrial taxa such as Staphylinidae and Hemiptera (Fig. 3a).

Nestedness analyses indicated that DS sediment invertebrate assemblages were not random, but nested along

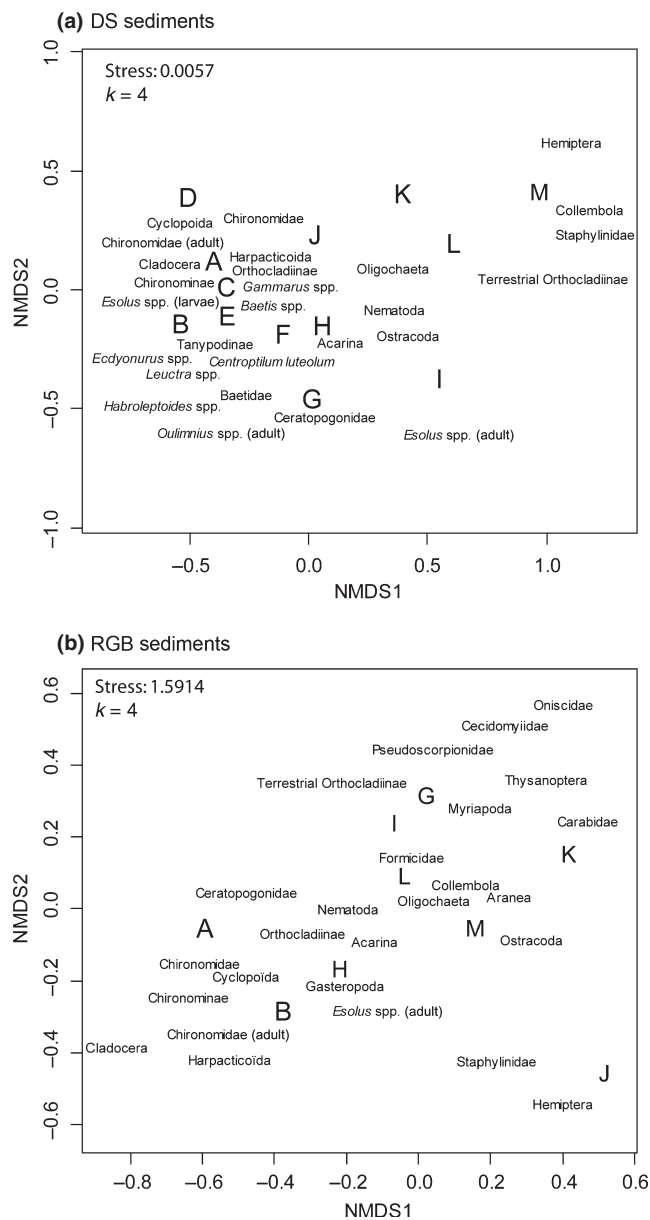


Fig. 3 Two-dimensional non-metric multidimensional scaling (NMDS) plots, based on log-transformed density data, of invertebrate assemblages recovered from rewetted sediments collected from (a) dry streambeds (DS) and (b) riparian gravel bars (RGB).

the gradient of increasing duration of the dry period ($T = 23.1$, Monte Carlo permutation test, $P < 0.001$). This indicated that invertebrate assemblages found at the most depauperate sites, those with the longest preceding dry period, were subsets of assemblages found at richest sites, which were those that had dried most recently.

Riparian gravel bar sediments. Aquatic taxon richness decreased steadily along the gradient of increasing duration of the dry period (Fig. 2b). On average, for every 20-

day increase in duration of the dry period, approximately one aquatic taxon was lost from the sediments. Semi-aquatic, aerial stages of aquatic taxa and terrestrial taxon richness did not show any relationship with duration of the dry (Fig. 2b). The overall density of aquatic taxa decreased steadily with increasing duration of the dry period, with an average of c. 12 individuals being lost from invertebrate samples for every 10-day increase in the duration of the dry period (Fig. 2b). Overall densities of semi-aquatic taxa, aerial stages of aquatic taxa and terrestrial taxa did not change with increasing duration of the dry period.

The NMDS ordination showed a poor separation of sites according to the length of the dry period (Fig. 3b). However, RGB sediments from sites with a preceding dry period <1 day (A and B) and site H (with a dry period duration of 25 days) were grouped together and had high densities of aquatic taxa, such as Cyclopoida, Harpacticoida, Cladocera or Diptera larvae (e.g. Ceratopogonidae, Chironomidae, Chironominae and Orthocladinae) (Fig. 3b). Sediments from gravel bars at other sites, with a preceding dry period duration >9 days, were on the right-hand side of the NMDS ordination and had high densities of terrestrial taxa, such as Oniscidae, Myriapoda, Pseudoscorpionidae, Aranea, Formicidae, Cecidomyiidae, Carabidae and Thysanoptera and high densities of semi-aquatic taxa such as Oligochaeta, Acarina, Nematoda and Collembola (Fig. 3b). Sediments from site J were isolated from the others sites, because of higher densities of terrestrial Staphylinidae and Hemiptera (Fig. 3b).

Riparian gravel bars sediment invertebrate assemblages were randomly distributed along the gradient of duration of the dry period and showed no nestedness ($T = 44.6$, Monte Carlo permutation test, $P = 0.65$).

Sediment moisture content. Sediment moisture content of DS sediments decreased very rapidly with increasing duration of the dry period (Fig. 4). Within the first 20 days of drying, sediment moisture content was more than halved, from 15 to 7%. In contrast, RGB sediment moisture content showed no relationship with increasing duration of the dry period (Fig. 4).

Convergence of DS and RGB invertebrate assemblages with increasing duration of the dry period

Similarity between DS and RGB invertebrate assemblages decreased steadily from 60 to 40% as the length of the dry period increased from 0.1 to 40 days (Fig 5). Between 40 and 50 days after drying, similarity between DS and RGB samples was halved. After 50 dry days, similarity between

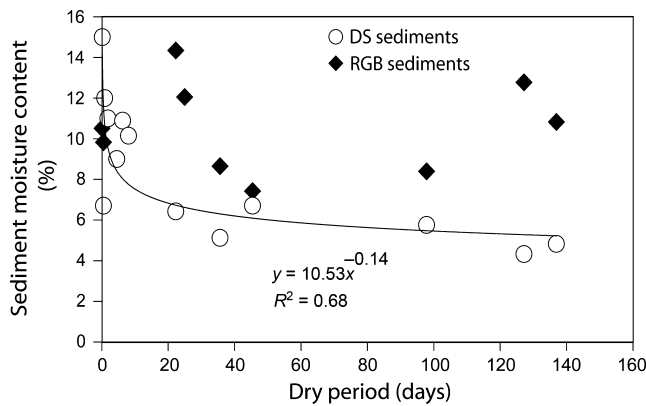


Fig. 4 Relationships between duration of the dry period and sediment moisture content for dry streambed (○) and riparian gravel bar (◆) sediments. The regression line fits duration of the dry period to DS sediment moisture content.

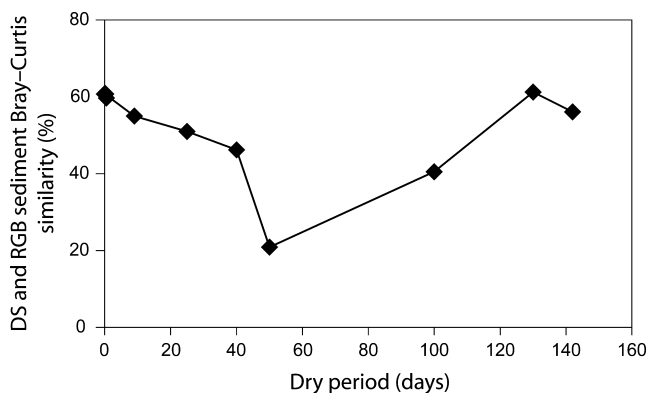


Fig. 5 Relationships between duration of the dry period and log-transformed density Bray-Curtis similarity between dry streambed and riparian gravel bar sediments.

sample types increased steadily to reach 60% after 130 days of drying.

Experimentally rewetted sediment and benthic invertebrates

A total of 33 taxa were found in pre-drying benthic samples, 31 in DS sediments and 47 in post-drying benthic samples (Appendix S1). Thirty-one taxa were common to pre- and post-drying benthic invertebrate assemblages, of which 20 taxa were also found in experimentally rewetted DS sediments. The taxa common to all sample types comprised Ephemeroptera (*Baetis* spp., *Ecdyonurus* spp., *Habroleptoides* spp., *Centroptilum luteolum*), Plecoptera (*Leuctra* spp.), Trichoptera (*Hydropsyche* spp.), Coleoptera (*Esolus* spp. and *Limnius* spp.), Diptera (Ceratopogonidae, Chironomidae, Chironominae, Psychodidae, Orthocla-diinae, Tanypodinae), *Gammarus* spp. (Amphipoda),

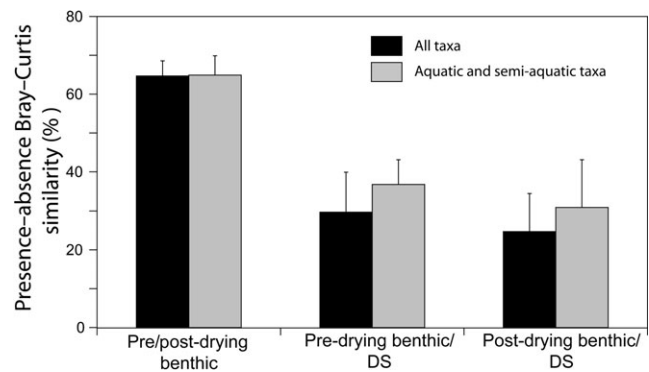


Fig. 6 Presence-absence Bray-Curtis similarities between pre- and post-drying benthic invertebrate assemblages and dry streambed invertebrate assemblages.

Cyclopoida, Harpacticoida, Ostracoda, Oligochaeta and Nematoda. Therefore, the number of aquatic taxa recovered from the experimentally rewetted DS sediments represented on average 65% of the taxa present in pre- and post-drying benthic. This proportion did not change substantially with an increasing dry period and ranged from 53 to 69% at the five sites considered, despite the fact that post-drying taxonomic richness and density of the benthos decreased as the preceding dry period increased ($R^2 = 0.49$ and 0.74 , respectively). In terms of density, the number of individuals of each of the 20 aquatic taxa recovered from rewetted sediment samples, and common to all sample types, ranged from one to 38 individuals, with an average of 7.25 L^{-1} of sediment.

Similarity analysis showed that pre- and post-drying benthic invertebrate samples were relatively similar in composition, with $60 \pm 4\%$ (mean $\pm$ SD) similarity (Fig. 6). Pre-drying benthic samples and DS sediments were $29 \pm 10\%$ similar, while DS sediments and post-drying benthic samples were $24 \pm 9\%$ similar (Fig. 6). These similarities between DS sediments and both pre- and post-drying samples did not change with increasing duration of dry period. In addition, when considering only aquatic and semi-aquatic taxa, these similarities increased only slightly (Fig. 6).

Discussion

Duration of the dry period

Until recently, dry sediments in temporary rivers were considered as 'biologically inactive' (Stanley, Fisher & Grimm, 1997), but several studies have refuted this assumption. In particular, some aquatic invertebrates can persist for several weeks or months in dry sediments and then recolonise benthic habitats after flow resumes

(Brock *et al.*, 2003; Tronstad *et al.*, 2005; Larned *et al.*, 2007; Stubbington *et al.*, 2009). Moreover, some terrestrial invertebrates from adjacent riparian zones colonise riverbeds and contribute to the biological activity of dry sections (Greenwood & McIntosh, 2010; Corti *et al.*, 2011). In our study, we simulated the onset of flow by artificially rewetting dry sediments that had previously been exposed to a wide range of dry periods. Our first prediction, that invertebrate responses to rewetting would vary with duration of the preceding dry period, was largely supported by the results. Artificial flooding of dry sediments led, within 32 days, to the appearance of aquatic invertebrate taxa in all samples. Since there was no possibility of aerial colonisation or drift, these aquatic invertebrates must have persisted as active or dormant (desiccation-resistant) life stages. In addition, terrestrial invertebrates were recovered from both rewetted DS and RGB sediments.

The taxon richness of aquatic organisms decreased linearly with increasing duration of the dry period in both DS and RGB sediments. In contrast, aquatic invertebrate density decreased as a power function of increasing dry period duration in DS sediments. This concurrent rapid decrease in density and linear decrease in richness suggest that many individuals from a small number of desiccation-sensitive taxa disappear soon after the temporary reach dries. However, a low-diversity assemblage of resistant taxa persists during prolonged dry periods, as reported by Larned *et al.* (2007) and Stubbington *et al.* (2009) in temporary rivers in New Zealand and the United Kingdom, respectively. Desiccation-sensitive taxa in the Albarine River included several Ephemeroptera (*Baetis* spp., *Centroptilum luteolum*, *Habroleptoides* spp.), Plecoptera (*Leuctra* spp.) and Coleoptera (*Esolus* spp., *Oulimnius* spp.). These taxa were present exclusively in sediments that had experienced very short (<7 days) dry periods, and thus remained relatively moist. These taxa may have some capacity for avoiding desiccation for short periods, such as crawling to moist benthic or hyporheic sediments. At the opposite end of the spectrum, desiccation-resistant taxa were those found in sediments after long (>4 weeks) dry periods and included the Cyclopoida, Ostracoda and Nematoda. Desiccation-resistant stages are well known in Cyclopoida and Ostracoda (e.g. Dahms, 1995; Dole-Olivier *et al.*, 2000), and the presence of copepodites in rewetted samples after long dry periods indicated hatching from desiccation-resistant eggs. Possible density-dependent emergence or mortality may occur during the initial phases of streambed drying, as indicated by the relationships between aquatic density and taxonomic richness with duration of

the dry period (Larned *et al.*, 2007). However, these processes are still poorly studied and require more experimental research.

These relationships with duration of the dry period have practical consequences for the management of temporary rivers. Indeed, artificial increases in the duration of the dry period, for example because of groundwater abstraction, surface water diversion or climate change, will inevitably reduce the richness and density of benthic invertebrate assemblages when flow resumes. In this study, every 10-day increase in the dry period removed approximately one taxon from the assemblage. This rate is much higher than that reported from a New Zealand temporary river (Selwyn River, with one taxon removed for every 50-day increase in the dry period, Larned *et al.*, 2007). However, the Albarine River seedbank was also much richer in taxa than that in the Selwyn River.

Two types of 'terrestrial' invertebrates were recovered from rewetted sediments: flying adults with aquatic larvae and riparian terrestrial taxa. Flying adults (Baetidae, Leuctridae and Chironomidae) were found exclusively in DS sediments and mostly after 8 days of rewetting, indicating they were not collected in the field as adults, but as larvae. Although they showed no relationships with the length of the dry period, most were collected in sediments with a dry period <50 days. Flying Ephemeroptera and Plecoptera clearly originated from the emergence of desiccation-sensitive Baetidae and Leuctridae, present in the sediments shortly after drying, and their late (October) emergence could be an adaptation of the life cycle to summer drying events. In contrast, adult Chironomidae were collected in sediments with longer dry periods (40–142 days) and may have originated from emergence of a few species with desiccation-resistant larvae (Tronstad *et al.*, 2005). The density of terrestrial invertebrates (Coleoptera, Orthocladinae, Hemiptera) increased with the duration of dry period in DS but not in RGB sediments, suggesting a progressive colonisation of the dry streambed (Corti *et al.*, 2011; Steward *et al.*, 2011). After a prolonged (142 days) dry period, however, the density of terrestrial taxa in DS sediments was still 20-fold lower than in adjacent RGB sediments. In addition, terrestrial invertebrates from RGB sediments showed little changes in composition with increasing duration of the dry period, suggesting that the presence or absence of aquatic prey in DS sediments was not an important driver of RGB terrestrial assemblages. Riparian predators may be able to modify their diet according to the prey types (aquatic or terrestrial) available. However, this requires confirmation using stable

isotopes (Paetzold *et al.*, 2005), for example, or gut content analysis (Hering & Plachter, 1997).

Convergence of DS and RGB invertebrate assemblages during dry phases

Invertebrate composition in DS and RGB sediments did not increase in similarity during dry phases. After 142 days of drying, DS sediments still harboured much lower density and richness of terrestrial organisms, and several riparian taxa found on RGB (e.g. Carabidae, Aranea, Oniscidae) were not recovered from DS sediments. Habitat characteristics may differ considerably between DS and RGB sediments; for instance, DS sediment moisture content was much lower than RGB sediments. This could be due to differences in streambed shading, sediment composition (Allen, 1970) or gravel bar vegetation (Brosfokske *et al.*, 1997; Chen *et al.*, 1999). However, these parameters are typically not measured, and this notion is speculative. In addition, DS assemblages were typically nested along the gradient of increasing duration of the dry period, indicating that invertebrate assemblages from sites experiencing the longest dry periods were subsets of assemblages from sites with shorter dry period durations. Such nestedness along the drying gradient has been reported previously (Arscott *et al.*, 2010; Datry, 2012) and seems to be common in temporary rivers, as drying often follows a longitudinal pattern (Larned *et al.*, 2011). However, no nestedness was detected in invertebrate assemblages from RGB sediments. These results suggest that temporal and spatial transition zones in temporary rivers have distinct biodiversity dynamics. The presence of spatial and temporal ecotones in river corridors increases habitat heterogeneity, which can increase biodiversity at the landscape scale (Ward & Tockner, 2001). In addition, longitudinal gradients of dry and wet periods in temporary rivers like the Albarine River increase successional diversity (i.e. there are mosaics of dry and saturated riverbed patches at various stages of terrestrial and aquatic succession) (Ward, Malard & Tockner, 2002; Larned *et al.*, 2007).

Experimentally rewetted sediment and benthic invertebrates

According to Lake (2003), resilience is a measure of the rate of recovery of a community following a disturbance. The survival of aquatic invertebrates in dry river sediments following the loss of surface water has been reported from temporary rivers located in different climatic zones (Boulton, 1989; Tronstad *et al.*, 2005; Larned *et al.*, 2007; Stubbington *et al.*, 2009). However, the contri-

bution of this mechanism to the recovery of benthic invertebrates was never quantified. In the Albarine River, more than half of the aquatic taxa found in the benthic zone both before and after summer drying events were also recovered from rewetted sediments, suggesting that the invertebrate seedbank could contribute substantially to the resilience of the benthos. Nevertheless, assemblages in before/after benthic samples and DS sediments were rather different, suggesting that recovery dynamics after rewetting were probably taxon specific and that other colonisation mechanisms have occurred. Although benthic invertebrate richness and density decrease with increasing duration of the dry period in the Albarine River (Datry, 2012), as in other temporary rivers (e.g. Datry, Larned & Scarsbrook, 2007; Arscott *et al.*, 2010), interestingly the proportion of common taxa found in DS sediments and in before/after benthic samples remained constant in the Albarine River across sites with different dry period duration. Therefore, the persistence of aquatic taxa in streambed sediments during dry phases could be an important means of maintaining aquatic invertebrate biodiversity in temporary rivers, particularly when streambed sediments remain sufficiently moist during dry periods. This conclusion highlights the importance of protecting dry reaches in temporary rivers, which are often impaired by recreation, sediment mining or sewage pollution (Larned *et al.*, 2010).

Additional processes must occur following flow resumption to explain the rapid recolonisation of rewetted benthic sediments by aquatic invertebrates. Ten taxa, representing 35% of benthic assemblage taxa, were not found in rewetted samples. Deeper hyporheic sediments are probably a poor refugium in the Albarine River during long dry periods, because they dry up completely after 5–15 days (Datry, 2012). Therefore, drift, aerial migration and oviposition require quantification following flow resumption to improve our understanding of the recolonisation dynamics of rewetted benthic habitats in temporary rivers (Williams & Hynes, 1976; Boulton, 1989; Stanley *et al.*, 1994).

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Supporting Information

Additional Supporting Information may be found in the online version of this article:

Appendix S1. List of the taxa with their classification as aquatic (A), semi-aquatic (S), aerial stage of aquatic taxa (AS) and terrestrial (T) recovered from rewetted sediment samples (dry streambed and riparian gravel bars) and pre- and post-drying benthic samples.

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